

GridBalance: Complete Technical Research Dossier

Exhaustive Repository Audit, Data Quality, ML Experiments, Robustness & Production Deployment

GridBalance — Final Retraining Decision Audit Report

Audit Status: Completed Evidence-Based Diagnostic

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1. Executive Summary & Final Retraining Decisions

- SGCC Electricity Theft Classification:** KEEP FROZEN (Classification: **A. DO NOT RETRAIN**)
 - Evidence:* The frozen 18-feature XGBoost model demonstrates healthy generalization with minimal performance decay between validation ($F1 = 0.4037$) and sealed test ($F1 = 0.3592$). It offers superior noise and missingness robustness compared to unreduced models.
 - Decision:* The frozen 18-feature model should remain the primary research classifier.
- UCI Electricity Load Forecasting:** KEEP FROZEN (Classification: **A. DO NOT RETRAIN**)
 - Evidence:* The frozen XGBoost regressor achieves an impressive $R^2 = 0.9942$ and $sMAPE = 1.981\%$ on the sealed 6-month test set (MAE = 4503.51 kWh).
 - Decision:* The frozen XGBoost regressor should remain the primary research forecasting model.

2. Quantitative Diagnostic Summary Matrix

Research Branch	Primary Model	Feature Space	Training Metric	Validation Metric	Sealed Test Metric	Generalization Assessment	Final Recommendation
SGCC Theft Detection	XGBoost Classifier	18 Features	F1 = 0.3874	F1 = 0.4037	F1 = 0.3592	Healthy Generalization	KEEP FROZEN
UCI Load Forecasting	XGBoost Regressor	29 Features	MAE = 2657.02 kWh	MAE = 3987.48 kWh	MAE = 4503.51 kWh	Excellent Generalization	KEEP FROZEN

3. Detailed Retraining Rationale

1. **Sealed Test Integrity:** The sealed test sets for both SGCC ($N=6,356$) and UCI ($N=4,416$) remain untouched for model selection decisions. Retraining or re-tuning hyperparameters without a new independent benchmark dataset risks invalidating the final test evaluation.
2. **Robustness & Complexity Tradeoff:** The 18-feature SGCC model matches full 42-feature performance while demonstrating lower prediction flip rates under severe telemetry degradation (20% Gaussian noise, 30% outage bursts).
3. **Training Diagnostic Saturation:** XGBoost logloss learning curves indicate that validation loss saturated near iteration 100 with early stopping active. Further boosting iterations would induce training set memorization without improving test generalization.